

4,212,2 Macroeconomics III

Exercises and Independent Studies

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Outline

- ▶ Endogenous Growth Model with Productive Externalities (Exercise 4.1)
- ▶ Endogenous Growth Model with R&D (Exercise 4.2)

Endogenous Growth Model with Productive Externalities:

Motivation

- ▶ The previous Solow models discussed in class had long-run economic growth driven by technological growth.
- ▶ However, technological growth was exogenous, $A_{t+1} = (1 + g)A_t$, and hence unexplained in these models.
- ▶ We further analyze different ways to endogenize the long-run technological growth rate. Therefore, we allow economic policy to influence long-run growth.
- ▶ One class of growth theories includes externalities in the accumulation of knowledge possessed by firms (physical capital) or by workers (human capital).
- ▶ The main idea is that a given firm is more productive the higher the average knowledge capital stock of other firms.

Endogenous Growth Model with Productive Externalities

- ▶ There is a large number of small firms, each with capital demand K_t^d , benefiting from the average stock of knowledge capital in the economy K_t in their goods production:

$$Y_t = (K_t^d)^\alpha (A_t L_t^d)^{1-\alpha}$$

- ▶ where A is taken as given, and there is CRS to (K, L) .
- ▶ Because of productive externalities, the individual firm's A_t depends on aggregate capital, K_t , i.e. $A_t = (K_t)^\phi$:

$$Y_t = (K_t^d)^\alpha (K_t)^{\phi(1-\alpha)} (L_t^d)^{1-\alpha}$$

Endogenous Growth Model with Productive Externalities

- ▶ In equilibrium, given that all firms are identical:

$$Y_t = (K_t)^{\alpha + \phi(1-\alpha)}(L_t)^{1-\alpha}$$

- ▶ The sum of the exponents is $1 + \phi(1 - \alpha)$ implying that there are IRS if $\phi > 0$.
- ▶ Hence, we have CRS at the individual firm level and IRS at the aggregate level.

Endogenous Growth Model with Productive Externalities

$$Y_t = (K_t)^\alpha (A_t L_t)^{1-\alpha},$$

$$A_t = (K_t)^\phi,$$

$$K_{t+1} = S_t + (1 - \delta)K_t,$$

$$w_t = (1 - \alpha) \left(\frac{K_t}{A_t L_t} \right)^\alpha A_t,$$

$$r_t = \alpha \left(\frac{K_t}{A_t L_t} \right)^{\alpha-1},$$

$$C_t + S_t = Y_t,$$

$$S_t = sY_t,$$

$$L_{t+1} = (1 + n)L_t.$$

Endogenous Growth Model with Productive Externalities

$$Y_t = (K_t)^\alpha ((K_t)^\phi L_t)^{1-\alpha}, \quad (1)$$

$$K_{t+1} = sY_t + (1 - \delta)K_t, \quad (2)$$

$$w_t = (1 - \alpha) \left(\frac{K_t}{(K_t)^\phi L_t} \right)^\alpha (K_t)^\phi, \quad (3)$$

$$r_t = \alpha \left(\frac{K_t}{(K_t)^\phi L_t} \right)^{\alpha-1}, \quad (4)$$

$$C_t = (1 - s)Y_t, \quad (5)$$

$$L_{t+1} = (1 + n)L_t. \quad (6)$$

5 Parameters: α , s , n , ϕ and δ . 6 Endogenous variables: Y , K , L , r , w , and C , of which K_t and L_t are state variables.

The Importance of ϕ

$$Y_t = (K_t)^{\alpha + \phi(1-\alpha)} (L_t)^{1-\alpha}$$

- ▶ If $\phi = 0$, then we have the basic Solow model
- ▶ If $\phi > 0$, then we have increasing returns in the aggregate production function
 - ▶ if $\phi < 1$, then $\alpha + \phi(1 - \alpha) < 1$: diminishing returns to the reproducible factor, capital, alone. This leads to “semi-endogenous growth”.
 - ▶ if $\phi = 1$, then $\alpha + \phi(1 - \alpha) = 1$: constant returns to capital alone. Leads to truly endogenous growth.

Semi-Endogenous Growth ($\phi < 1$)

Transformations in per 'Efficiency Unit of Labor' Terms

Define: $\tilde{k}_t = \frac{k_t}{A_t} = \frac{K_t}{A_t L_t}$ and $\tilde{y}_t = \frac{y_t}{A_t} = \frac{Y_t}{A_t L_t}$. Then:

$$\tilde{y}_t = \left(\frac{K_t}{A_t L_t} \right)^\alpha \left(\frac{A_t L_t}{A_t L_t} \right)^{1-\alpha} = \tilde{k}_t^\alpha.$$

From $A_t = (K_t)^\phi$, we can derive $\frac{A_{t+1}}{A_t} = \left(\frac{K_{t+1}}{K_t} \right)^\phi$. We further use this relationship to derive:

$$\begin{aligned} \frac{\tilde{k}_{t+1}}{\tilde{k}_t} &= \frac{K_{t+1}/K_t}{(A_{t+1}/A_t)(L_{t+1}/L_t)} \\ &= \frac{K_{t+1}/K_t}{(K_{t+1})^\phi / (K_t)^\phi (1+n)} = \frac{1}{(1+n)} \left(\frac{K_{t+1}}{K_t} \right)^{1-\phi} \end{aligned}$$

Semi-Endogenous Growth ($\phi < 1$)

We further insert the law of motion for capital

$$K_{t+1} = sY_t + (1 - \delta)K_t:$$

$$\begin{aligned}\frac{\tilde{k}_{t+1}}{\tilde{k}_t} &= \frac{1}{(1+n)} \left(\frac{sY_t + (1-\delta)K_t}{K_t} \right)^{1-\phi} \\ &= \frac{1}{(1+n)} \left(s\frac{Y_t}{K_t} + (1-\delta) \right)^{1-\phi} \\ &= \frac{1}{(1+n)} \left(s\frac{\tilde{y}_t}{\tilde{k}_t} + (1-\delta) \right)^{1-\phi} \\ &= \frac{1}{(1+n)} \left(s\tilde{k}_t^{\alpha-1} + (1-\delta) \right)^{1-\phi}\end{aligned}$$

Semi-Endogenous Growth ($\phi < 1$)

Rearranging the terms delivers the transition equation:

$$\begin{aligned}\tilde{k}_{t+1} &= \frac{\tilde{k}_t}{(1+n)} \left(s\tilde{k}_t^{\alpha-1} + (1-\delta) \right)^{1-\phi} \\ &= \frac{\tilde{k}_t^{1-\phi+\phi}}{(1+n)} \left(s\tilde{k}_t^{\alpha-1} + (1-\delta) \right)^{1-\phi} \\ &= \frac{\tilde{k}_t^\phi}{(1+n)} \left(s\tilde{k}_t^\alpha + (1-\delta)\tilde{k}_t \right)^{1-\phi}\end{aligned}$$

In the lecture notes:

$$\tilde{k}_{t+1} = \frac{1}{(1+n)} \left(s\tilde{k}_t^{\frac{\alpha+\phi-\alpha\phi}{1-\phi}} + (1-\delta)\tilde{k}_t^{\frac{1}{1-\phi}} \right)^{1-\phi}$$

Steady State

Set $\tilde{k}_{t+1} = \tilde{k}_t = \tilde{k}^*$. This implies:

$$\frac{1}{(1+n)} \left(s\tilde{k}_t^{\alpha-1} + (1-\delta) \right)^{1-\phi} = 1,$$

which can be solved to obtain the steady state value $\tilde{k}^*$:

$$\tilde{k}^* = \left(\frac{s}{(1+n)^{\frac{1}{(1-\phi)}} - (1-\delta)} \right)^{\frac{1}{(1-\alpha)}} > 0$$

if $n + \delta > 0$.

Steady State

We have derived that:

$$\tilde{y}_t = \tilde{k}_t^\alpha,$$

which implies that the steady state value $\tilde{y}^*$ is given by:

$$\tilde{y}^* = (\tilde{k}^*)^\alpha = \left(\frac{s}{(1+n)^{\frac{1}{1-\phi}} - (1-\delta)} \right)^{\frac{\alpha}{1-\alpha}}$$

You can further use these results to obtain the steady-state values of the other model variables $\tilde{c}^*$, $\tilde{w}^*$ and r^* .

Variables in per 'Worker' Terms

$$k_t = \tilde{k}_t A_t$$

$$y_t = \tilde{y}_t A_t$$

However, as opposed to previous models where the growth rate was exogenous, $A_t = (1 + g)^t A_0$, here we have endogenous growth and we have to derive its steady-state value.

We derived earlier that:

$$\frac{\tilde{k}_{t+1}}{\tilde{k}_t} = \frac{1}{(1+n)} \left(\frac{K_{t+1}}{K_t} \right)^{1-\phi}$$

Using that in the steady state $\tilde{k}_{t+1} = \tilde{k}_t = \tilde{k}^*$, we find that:

$$1 = \frac{1}{(1+n)} \left(\frac{K_{t+1}}{K_t} \right)^{1-\phi}$$

Variables in per 'Worker' Terms

Rearranging terms gives:

$$\left(\frac{K_{t+1}}{K_t} \right) = (1 + n)^{\frac{1}{(1-\phi)}}$$

Use that:

$$\frac{A_{t+1}}{A_t} = \left(\frac{K_{t+1}}{K_t} \right)^{\phi} = (1 + n)^{\frac{\phi}{(1-\phi)}},$$

which implies that the steady-state growth of the economy is given by:

$$\frac{A_{t+1} - A_t}{A_t} = \frac{A_{t+1}}{A_t} - 1 = (1 + n)^{\frac{\phi}{(1-\phi)}} - 1$$

Variables in per 'Worker' Terms

$$g_{se} = (1 + n)^{\frac{\phi}{(1-\phi)}} - 1$$

- ▶ How does the growth rate depend on the model parameters n and ϕ ?
- ▶ When $\phi < 1$ the exponent on $1 + n$ is positive. Hence, a larger population growth rate gives a higher growth rate of output per worker in the steady state. Not realistic!
- ▶ A larger ϕ means a larger impact of the productive externality, reducing the degree of diminishing returns to capital (and to capital per worker). Diminishing returns to capital cause convergence, so as diminishing returns disappear, convergence becomes increasingly slow. When $\phi = 1$, the productive externality is strong enough to eliminate diminishing returns, implying no convergence at all.

Increase in the Savings Rate

$$g_{se} = (1 + n)^{\frac{\phi}{(1-\phi)}} - 1$$

- ▶ What is the impact of the increase in the savings rate on the steady-state growth rate g_{se} ?
- ▶ None
- ▶ What is the impact on the steady-state levels?

$$\tilde{k}^* = \left(\frac{s}{(1 + n)^{\frac{1}{(1-\phi)}} - (1 - \delta)} \right)^{\frac{1}{(1-\alpha)}}$$

- ▶ An increase in the savings rate leads to more steady-state capital and output.

Increase in the Savings Rate

- ▶ Intuitively, is the impact on the steady-state levels larger or smaller in this model compared to the Solow model with exogenous growth?
- ▶ The same for steady-state variables in per efficiency units of labor terms (check by taking the derivative w.r.t. s).
- ▶ Semi-Endogenous Growth Model ($\phi < 1$)

$$\tilde{k}^* = \left(\frac{s}{(1+n)^{\frac{1}{(1-\phi)}} - (1-\delta)} \right)^{\frac{1}{(1-\alpha)}}$$

- ▶ General Solow Model:

$$\tilde{k}^* = \left(\frac{s}{n + g + \delta + ng} \right)^{\frac{1}{(1-\alpha)}}$$

Increase in the Savings Rate

- ▶ Intuitively, is the impact on the steady-state levels larger or smaller in this model compared to the Solow model with exogenous growth?
- ▶ Higher for the steady-state growth path of variables.
- ▶ Semi-Endogenous Growth Model ($\phi < 1$)

$$k_t^* = \tilde{k}^* A_t = \tilde{k}^* (K_t)^\phi = \tilde{k}^* (k_t^* L_t)^\phi = (\tilde{k}^*)^{\frac{1}{1-\phi}} (L_t)^{\frac{\phi}{1-\phi}}$$

- ▶ General Solow Model:

$$k_t^* = \tilde{k}^* A_t = \tilde{k}^* (1 + g)^t A_0$$

Recovering Remaining Model Variables

$$\tilde{k}_{t+1} = \frac{\tilde{k}_t^\phi}{(1+n)} \left(s\tilde{k}_t^\alpha + (1-\delta)\tilde{k}_t \right)^{1-\phi}$$

$$\tilde{y}_t = \tilde{k}_t^\alpha$$

$$\tilde{c}_t = (1-s)\tilde{k}_t^\alpha$$

$$L_{t+1} = (1+n)L_t$$

$$K_t = (\tilde{k}_t L_t)^{\frac{1}{(1-\phi)}}$$

$$A_t = (\tilde{k}_t L_t)^{\frac{\phi}{(1-\phi)}}$$

$$\ln(y_t) = \ln(\tilde{y}_t) + \ln(A_t)$$

$$\ln(k_t) = \ln(\tilde{k}_t) + \ln(A_t)$$

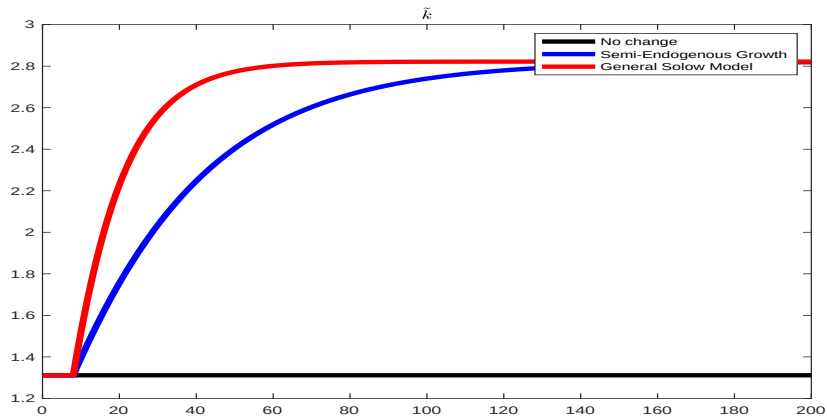
$$\ln(c_t) = \ln(\tilde{c}_t) + \ln(A_t)$$

$$g_t^y = \ln(y_t) - \ln(y_{t-1})$$

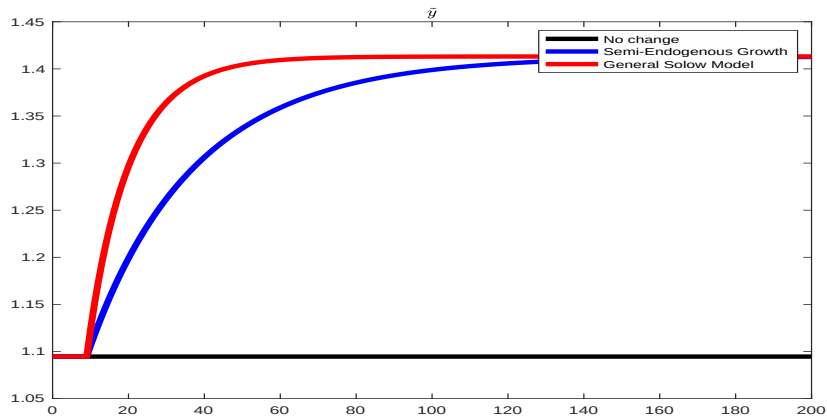
Simulations

- ▶ Calibration: $\alpha = 0.33$, $\phi = 0.6$, $\delta = 0.1$, $s = 0.15$, $n = 0.01$, $L_0 = 1$.
- ▶ What happens if the savings rate s increases from 0.15 to 0.25 in period 10?

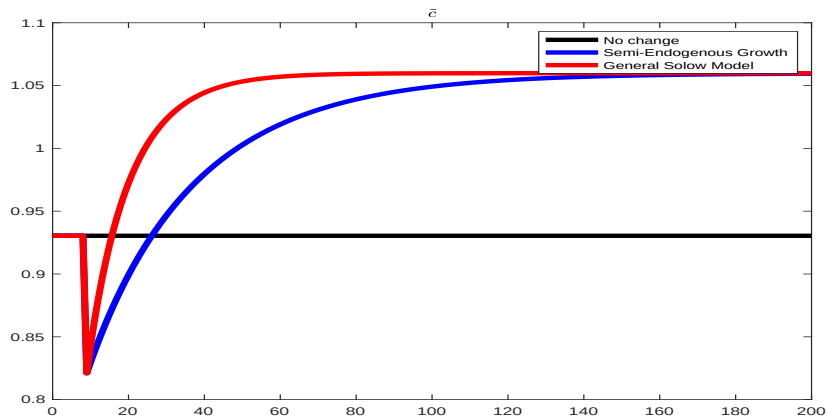
Increase in the Savings Rate



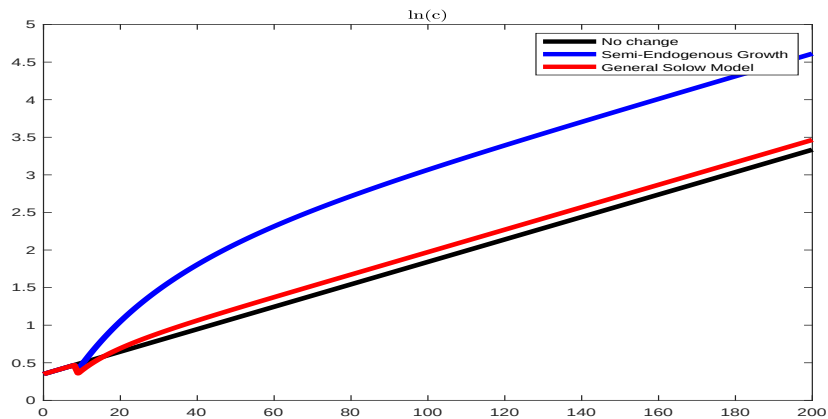
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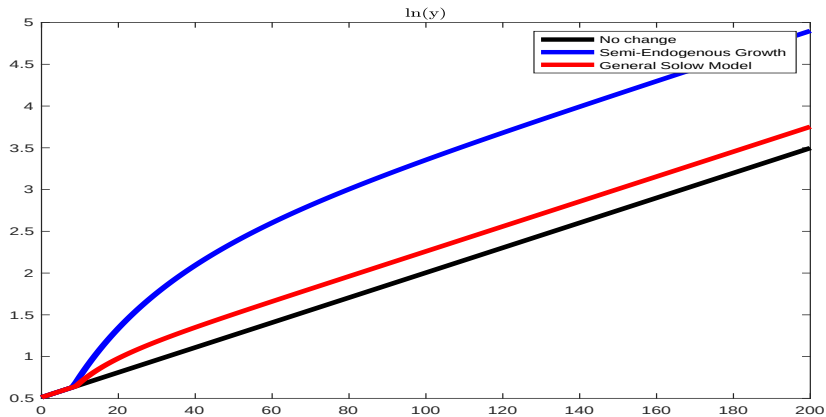
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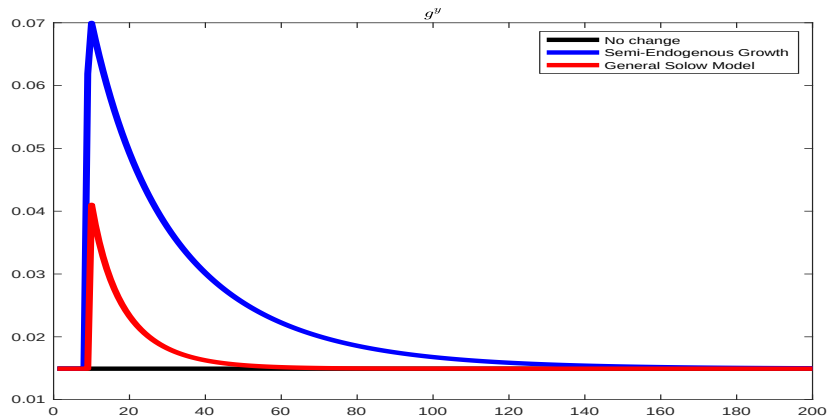
Increase in the Savings Rate



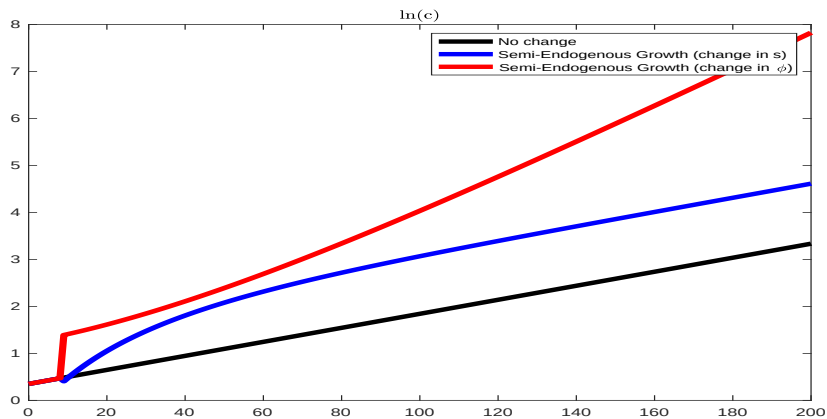
Increase in the Savings Rate



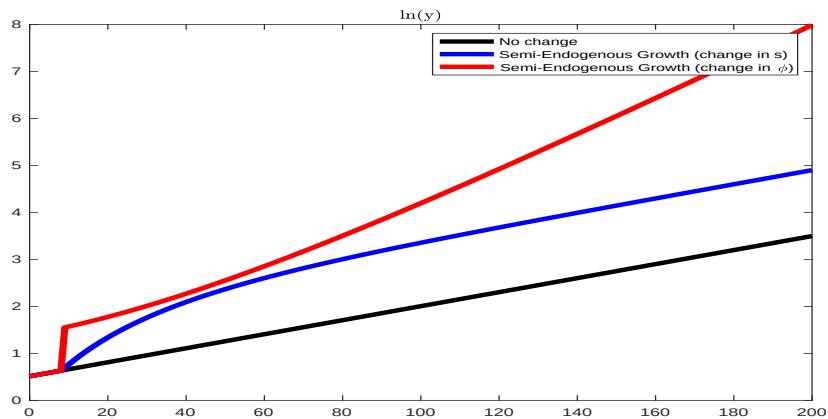
Increase in the Savings Rate



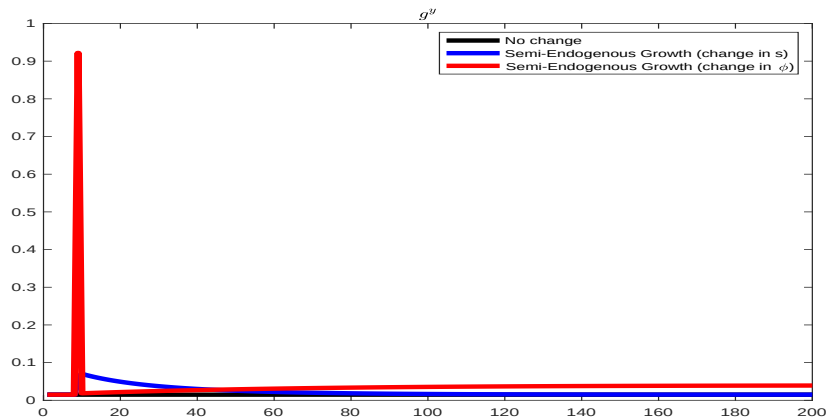
Increase in ϕ



Increase in ϕ



Increase in ϕ



Simulations for $\phi = 1$

- Now assume that $\phi = 1$ and $n = 0$. Hence the calibration becomes: $\alpha = 0.33$, $\phi = 1$, $n = 0$, $\delta = 0.1$, $s = 0.15$, $L_0 = 1$, $k_0 = 1$.

Endogenous Growth ($\phi = 1$): The AK Model

$$Y_t = AK_t,$$

where

$$A = L^{1-\alpha},$$

and the law of motion of capital is as usual,

$K_{t+1} = sY_t + (1 - \delta)K_t$. We further divide through by L and combine the two equations to obtain the transition equation:

$$k_{t+1} = sAk_t + (1 - \delta)k_t,$$

which we can transform to derive the growth rate:

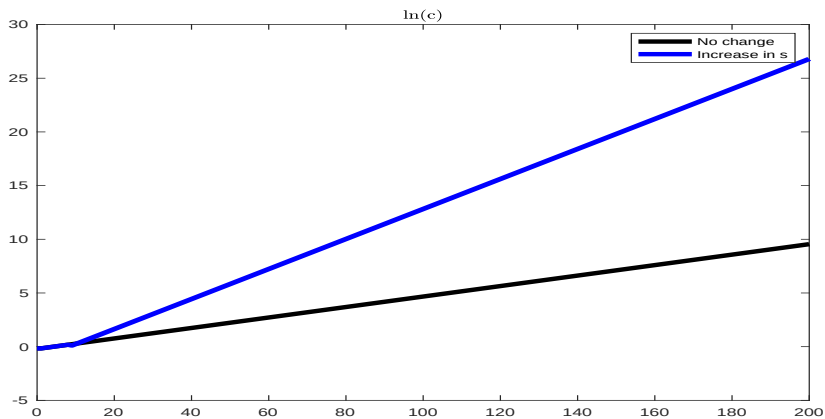
$$\frac{k_{t+1} - k_t}{k_t} = sA - \delta$$

Endogenous Growth ($\phi = 1$): The AK Model

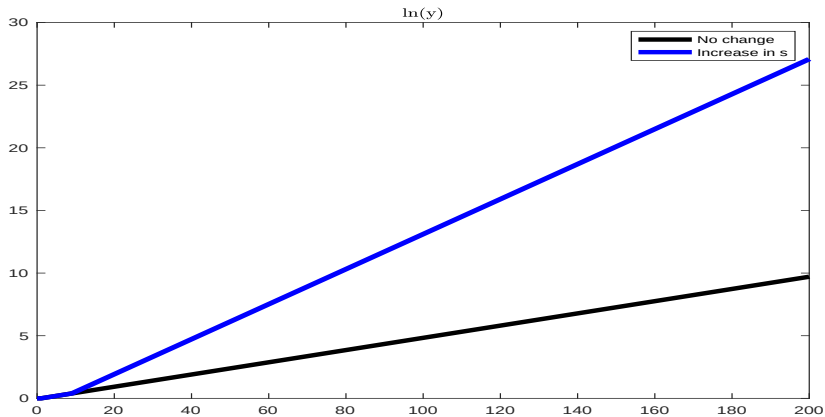
$$\frac{k_{t+1} - k_t}{k_t} = sA - \delta$$

- ▶ What is the effect of the change in the savings rate on the growth rate here?
- ▶ An increase in s gives a higher growth rate of output per worker.
- ▶ Truly endogenous growth: positive growth in output without exogenous technological growth or population growth; the growth rate is given by model parameters, among these s .
- ▶ Structural policies: higher s and lower δ give everlasting higher growth in GDP and consumption per capita.
- ▶ Critique - Scale effect: remember that $A = L^{1-\alpha}$. A larger constant population results in higher growth and an increasing population gives accelerating growth. Not realistic!

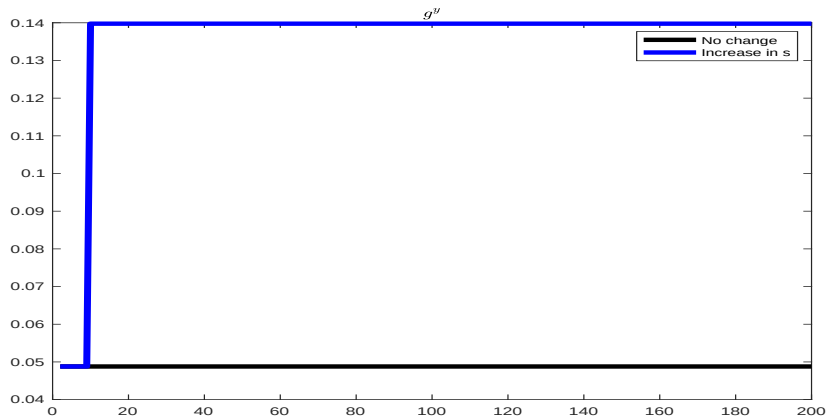
Increase in the Savings Rate



Increase in the Savings Rate



Increase in the Savings Rate



Endogenous Growth Model with R&D: Motivation

- ▶ So far technological progress has been either:
 - ▶ Unexplained (exogenous growth: Chapters 5, 6 and 7) or
 - ▶ Unintentional (endogenous growth model based on productive externalities: Chapter 8).
- ▶ These models do not account for the importance of R&D in driving technological progress.
- ▶ However, both firms and governments worldwide invest considerably in R&D and many people work in this sector.
- ▶ We further analyze the class of growth theories in which technological progress is the endogenous and intentional result of an R&D production process.
- ▶ The main idea is that the R&D sector hires labor to produce ideas, which ultimately lead to a higher level of technology.

Endogenous Growth Model with R&D

- ▶ There are two production sectors, their output being final goods and new technology.
- ▶ The final goods sector:

$$Y_t = (K_t)^\alpha (A_t L_{Y_t})^{1-\alpha}$$

- ▶ where A_t is taken as given and is the productive effect of the stock of all previous ideas.
- ▶ All firms can use A_t simultaneously because ideas are non-rival.
- ▶ There is CRS to (K, L) , but IRS to (K, L, A) .

Endogenous Growth Model with R&D

- ▶ The R&D sector is populated by a large number of small firms, which together produce the aggregate output of new technology, $A_{t+1} - A_t$.
- ▶ Each R&D firm takes total the labour input, L_{At} , the aggregate production, $A_{t+1} - A_t$, and the aggregate stock of knowledge, A_t , as given.
- ▶ The individual firm's production function is:

$$a_t = \bar{\rho} A_t^\phi L_{At}^d, \bar{\rho} > 0,$$

- ▶ At aggregate level:

$$\bar{\rho} = \rho L_{At}^{\lambda-1}, a_t = A_{t+1} - A_t, L_{At}^d = L_{At}.$$

- ▶ Combining these equations leads to the aggregate production function of the R&D sector:

$$A_{t+1} - A_t = \rho A_t^\phi L_{At}^\lambda.$$

Endogenous Growth Model with R&D

$$Y_t = (K_t)^\alpha (A_t L_{Yt})^{1-\alpha},$$

$$A_{t+1} - A_t = \rho A_t^\phi L_{At}^\lambda,$$

$$K_{t+1} = S_t + (1 - \delta)K_t,$$

$$L_{t+1} = (1 + n)L_t,$$

$$L_{Yt} + L_{At} = L_t,$$

$$L_{At} = s_R L_t,$$

$$C_t + S_t = Y_t,$$

$$S_t = s Y_t.$$

8 Parameters: $\alpha, s, s_R, n, \phi, \rho, \lambda$ and δ . 8 Endogenous variables: Y, K, L, L_A, L_Y, S, C , and A , of which K_t, A_t and L_t are state (predetermined) variables.

The First Law of Motion

- ▶ The growth rate of technology:

$$g_t = \frac{A_{t+1} - A_t}{A_t} = \rho A_t^{\phi-1} L_{A_t}^\lambda = \rho A_t^{\phi-1} (s_R L_t)^\lambda$$

- ▶ The dynamic evolution of the growth rate of technology:

$$g_{t+1} = (1 + g_t)^{\phi-1} g_t (1 + n)^\lambda$$

- ▶ From any initial value g_0 , the growth rate of technology evolved independently of other endogenous variables as it only depends on parameters.

The Second Law of Motion

Transformations in per 'Efficiency Unit of Labor' Terms

Define: $\tilde{k}_t = \frac{k_t}{A_t} = \frac{K_t}{A_t L_t}$ and $\tilde{y}_t = \frac{y_t}{A_t} = \frac{Y_t}{A_t L_t}$. Then:

$$\tilde{y}_t = \left(\frac{K_t}{A_t L_t} \right)^\alpha \left(\frac{A_t(1-s_R)L_t}{A_t L_t} \right)^{1-\alpha} = \tilde{k}_t^\alpha (1-s_R)^{1-\alpha}.$$

From the law of motion for capital $K_{t+1} = sY_t + (1-\delta)K_t$, we can derive:

$$\tilde{k}_{t+1} = \frac{s\tilde{k}_t^\alpha (1-s_R)^{1-\alpha} + (1-\delta)\tilde{k}_t}{(1+n)(1+g_t)}$$

Note that to compute $\tilde{k}_{t+1}$, you need to know not only $\tilde{k}_t$ but also g_t .

The Importance of ϕ

$$A_{t+1} - A_t = \rho A_t^\phi L_{At}^\lambda$$

- ▶ Is strictly positive as long as $L_{At} > 0$, implying that $g_t > 0$. Knowledge is accumulated over time.
- ▶ Assume that we have constant labor input in R&D, $L_{At} = L_A$, then $A_{t+1} - A_t = \rho A_t^\phi L_A^\lambda$ and $g_t = \rho A_t^{\phi-1} L_A^\lambda$:
 - ▶ if $\phi = 1$, then $g_t = \rho L_A^\lambda$: constant labor input in R&D maintains a constant growth rate of A_t . Leads to strictly endogenous growth.
 - ▶ if $0 < \phi < 1$, then $g_t \rightarrow 0$ as $t \rightarrow \infty$ because $A_t^{\phi-1} \rightarrow 0$: constant labor input in R&D does not maintain a constant growth rate of A_t . For this, we need a growing labor input. This leads to “semi-endogenous growth”.

Semi-Endogenous Growth ($\phi < 1$)

Steady State

Define: $g_{t+1} = g_t = g_{se}$. Then:

$$g_{se} = (1 + n)^{\frac{\lambda}{1-\phi}} - 1$$

Set $\tilde{k}_{t+1} = \tilde{k}_t = \tilde{k}^*$. Replace g_t with g_{se} in the law of motion for capital. This implies:

$$\tilde{k}^* = \left(\frac{s}{n + g_{se} + \delta + ng_{se}} \right)^{\frac{1}{(1-\alpha)}} (1 - s_R)$$

We then have that:

$$\tilde{y}^* = (\tilde{k}^*)^\alpha (1 - s_R)^{1-\alpha} = \left(\frac{s}{n + g_{se} + \delta + ng_{se}} \right)^{\frac{\alpha}{(1-\alpha)}} (1 - s_R)$$

$$\tilde{c}^* = (1 - s)\tilde{y}^* = (1 - s) \left(\frac{s}{n + g_{se} + \delta + ng_{se}} \right)^{\frac{\alpha}{(1-\alpha)}} (1 - s_R)$$

Variables in per 'Worker' Terms

$$k_t = \tilde{k}_t A_t$$

$$y_t = \tilde{y}_t A_t$$

$$c_t = \tilde{c}_t A_t$$

On the balanced growth path the growth rate of technology is constant, $\rho A_t^{\phi-1} L_{At}^\lambda = g_{se}$. This implies that:

$$A_t = \left(\frac{\rho (s_R L_t)^\lambda}{g_{se}} \right)^{\frac{1}{1-\phi}} = \left(\frac{\rho}{g_{se}} \right)^{\frac{1}{1-\phi}} s_R^{\frac{\lambda}{1-\phi}} L_t^{\frac{\lambda}{1-\phi}}$$

Use that:

$$L_t^{\frac{\lambda}{1-\phi}} = (L_0(1+n)^t)^{\frac{\lambda}{1-\phi}} = L_0^{\frac{\lambda}{1-\phi}} (1+g_{se})^t$$

Variables in per 'Worker' Terms

Steady state growth path:

$$k_t^* = \tilde{k}^* A_t$$

$$y_t^* = \tilde{y}^* A_t$$

$$c_t^* = \tilde{c}^* A_t$$

Hence, c_t^* is:

$$\begin{aligned} c_t^* &= \tilde{c}^* A_t = (1-s) \left(\frac{s}{n + g_{se} + \delta + ng_{se}} \right)^{\frac{\alpha}{(1-\alpha)}} (1-s_R) A_t \\ &= (1-s) \left(\frac{s}{n + g_{se} + \delta + ng_{se}} \right)^{\frac{\alpha}{1-\alpha}} (1-s_R) s_R^{\frac{\lambda}{1-\phi}} \left(\frac{\rho}{g_{se}} \right)^{\frac{1}{1-\phi}} L_0^{\frac{\lambda}{1-\phi}} (1+g_{se})^t, \end{aligned}$$

where $g_{se} = (1+n)^{\frac{\lambda}{1-\phi}} - 1$.

Golden Rules

To derive the golden rules of s and s_R which maximize c_t^* , you need to take the first order conditions of c_t^* with respect to s and s_R .

For simplicity, you could first take logs of c_t^* and use that if

$f(x) = \ln(g(x))$ then $f'(x) = \frac{g'(x)}{g(x)}$:

$$\frac{\partial \ln c_t^*}{\partial s} = 0 \rightarrow -\frac{1}{1-s} + \frac{\alpha}{1-\alpha} \frac{1}{s} = 0 \rightarrow s = \alpha$$

$$\frac{\partial \ln c_t^*}{\partial s_R} = 0 \rightarrow -\frac{1}{1-s_R} + \frac{\lambda}{1-\phi} \frac{1}{s_R} = 0 \rightarrow s_R = \frac{\lambda}{1-\phi + \lambda}$$

Golden Rule s_R

$$s_R = \frac{\lambda}{1 - \phi + \lambda}$$

- ▶ How does the golden rule s_R depend on the model parameters λ and ϕ ?
- ▶ The research labor share, s_R , affects the growth path of c_t^* through two opposite effects: A larger s_R means that a larger fraction of the labor force is employed in the R&D sector, so less labor is allocated to the production of final goods pulling y_t^* and c_t^* downwards, but also the technological level A_t along the steady-state growth path will be higher, pulling y_t^* and c_t^* upwards.
- ▶ s_R depends positively on both λ and ϕ .
- ▶ Larger λ implies less negative “stepping on toes” effect.
- ▶ Larger ϕ (but $\phi < 1$) implies a stronger “standing on shoulders” effect and/or a weaker “fishing out” effect.

Steady State Growth Rate

- Explain why $n > 0$ does not imply increasing (but rather positive and constant) growth rates in the long run.

$$g_{se} = (1 + n)^{\frac{\lambda}{1-\phi}} - 1$$

- In the model of semi-endogenous where $0 < \phi < 1$, the technological and economic growth rates are constant and positive in the long run only if $n > 0$.
- Intuitively, when $\phi < 1$, a constant labor input will imply that the growth rate will go to zero in the long run. Hence, to have a positive growth rate in A_t , more labor in research is needed, and having s_R constant, this can only be achieved with an increasing labor force.

The 'Scale Effect'

$$y_t^* = \tilde{y}^* A_t$$

Hence, y_t^* is:

$$\begin{aligned} y_t^* = \tilde{y}^* A_t &= \left(\frac{s}{n + g_{se} + \delta + n g_{se}} \right)^{\frac{\alpha}{(1-\alpha)}} (1 - s_R) A_t \\ &= \left(\frac{s}{n + g_{se} + \delta + n g_{se}} \right)^{\frac{\alpha}{1-\alpha}} (1 - s_R) s_R^{\frac{\lambda}{1-\phi}} \left(\frac{\rho}{g_{se}} \right)^{\frac{1}{1-\phi}} L_0^{\frac{\lambda}{1-\phi}} (1 + g_{se})^t, \end{aligned}$$

where $g_{se} = (1 + n)^{\frac{\lambda}{1-\phi}} - 1$.

- ▶ A larger initial labor force, L_0 , implies a higher steady state growth path, y_t^* : more workers imply a higher level of income per worker in the long run.

The 'Scale Effect'

- ▶ The intuition for the level scale effect is just that a larger labor force in all periods as implied by a larger L_0 , for a given research share s_R implies a higher labor input into the R&D sector in all periods bringing a higher level of technology, A_t , in all periods.
- ▶ Note that nothing similar is true in the Solow models with exogenous technological progress (Chapters 5, 6, and 7), whereas the semi-endogenous growth model of Chapter 8 had a similar level scale effect.